

Software Requirements Specification

EV Charging Station Reservation System

Problem Statement #21

Document	Details
Prepared by	Shriya S R
SRN	PES1UG24CS449
Class and Section	5 H
Course	Software Engineering
Version	1.0
Date	02 October 2026
Status	Draft for submission
Standard followed	Structure based on IEEE 830 / ISO/IEC/IEEE 29148

Note on sources. The seven requirements (FR001 to FR005, NFR001, NFR002), their priorities, story points, and the seven reported defects are taken from the Jira Scrum, Kanban and Bug Tracking spaces created for this problem statement. Where the problem statement does not give a value (for example response-time targets), the value is marked **(assumed)** so it can be confirmed or changed.

Table of Contents

1. Introduction
2. Overall Description
3. Specific Requirements
4. Requirements Traceability
5. Appendix A: Jira Work Item Mapping
6. Appendix B: Revision History

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements of the EV Charging Station Reservation System. It is intended for the developer, the tester, and the course evaluators, and is the basis for design, coding, testing and the Requirements Traceability Matrix (RTM).

1.2 Scope

The system lets an electric vehicle (EV) driver find charging stations, reserve a charging slot on a charger that matches the vehicle connector, see the dynamic charging rate, and cancel a reservation. It keeps charger availability up to date and automatically frees a reserved socket when the driver does not show up. Billing hardware, the physical chargers and the payment gateway are outside the scope of this system and are treated as external systems.

1.3 Definitions, Acronyms and Abbreviations

Term	Meaning
EV	Electric Vehicle
FR / NFR	Functional Requirement / Non-Functional Requirement
SRS	Software Requirements Specification
RTM	Requirements Traceability Matrix
Connector type	The plug standard of a vehicle or charger (for example Type 2, CCS, CHAdeMO)
Slot	A time window on a specific charger that can be reserved
Socket	The physical charging outlet on a charger
No-show	A reservation where the driver does not arrive or start charging within the grace period
SP	Story points used for estimation in Jira

1.4 References

- 21_SE_Lab1_SE_Problem_Statements.pdf (Problem Statement #21)
- Jira spaces: Scrum_Banana_Problem#21, Kanban_Problemstatement#21, Bugtracking_problemstatemet#21
- Lab 1 artefacts: SE Lab 1 UML, Alternate flow diagram, Exception flow diagram and document
- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications

1.5 Document Overview

Section 2 describes the product context, users and constraints. Section 3 lists each requirement with acceptance criteria. Section 4 traces requirements to Jira stories, tasks and defects. Appendix A maps Jira IDs.

2. Overall Description

2.1 Product Perspective

The system is a stand-alone reservation service that talks to three external parties: the EV driver (through a web or mobile interface), the charging stations (to read and update charger status), and a payment gateway (to confirm payment before a reservation is confirmed).

2.2 Product Functions (summary)

- Match vehicle connector type with compatible chargers and reserve an available slot (FR001)
- Search charging stations and filter by compatibility and availability (FR002)
- Calculate and display the dynamic charging rate (FR003)
- Cancel a reservation and release the slot (FR004)
- Manage and synchronise charger status (FR005)
- Detect no-shows and release the socket automatically (NFR001)
- Stay responsive and scale under peak load (NFR002)

2.3 User Classes and Characteristics

User class	Description	Main needs
EV Driver	Primary user. Basic app skills, wants quick and reliable reservation.	Search, reserve, see price, cancel
Station Operator / Admin	Manages chargers and their status.	Update status, keep availability correct
Payment Gateway (external)	Third-party system that processes payment.	Payment success or failure response

2.4 Operating Environment

Web application accessible from a modern browser (and a mobile-friendly interface), with a server-side application and a database. Exact technology stack is **(assumed)** to be decided during design.

2.5 Design and Implementation Constraints

- Reservations may be confirmed only for slots that are available at the requested time.
- Connector compatibility must be checked before a reservation is created.
- Charger status must be consistent across search, reservation and cancellation.

2.6 Assumptions and Dependencies

- A payment gateway is available and returns a clear success or failure response **(assumed, from defect BP-6)**.
- Chargers report or accept status updates through a defined interface.
- Drivers have an account and have saved the connector type of their vehicle.
- The no-show grace period is 10 minutes (from defect BP-7).

3. Specific Requirements

3.1 External Interface Requirements

Interface	Requirement
User interface	Screens for search, charger details with rate, reservation, cancellation and reservation history. Error messages must state the reason (for example incompatible connector or slot unavailable).
Charger / station interface	The system reads charger status (available, reserved, occupied, out of service) and updates it on reservation, cancellation, no-show release and charging start or end.
Payment interface	The system sends the payable amount and receives success or failure. A reservation is confirmed only on success.

3.2 Functional Requirements

FR001: Compatible Charger Reservation

Priority	Jira reference	Story
High	SBP-1 / KP-14, 8 SP	As an EV driver, I want the system to match my vehicle connector type with compatible charging stations and reserve an available charging slot, so that I can reliably charge my vehicle using a suitable charger.

Requirements

- **FR001.1** The system shall let the driver select or use the saved connector type of the vehicle.
- **FR001.2** The system shall offer only chargers whose connector type is compatible with the vehicle.
- **FR001.3** The system shall create a reservation only for a slot that is available for the requested time.
- **FR001.4** The system shall reject the request and show the reason if the slot is unavailable or the connector is incompatible.
- **FR001.5** The system shall confirm the reservation only after payment succeeds (**assumed**).

Acceptance criteria

- A driver with a Type 2 vehicle is never offered a CCS-only charger (defect BP-1).
- Booking an already reserved slot is refused (defect BP-2).
- If payment succeeds, the reservation status becomes Confirmed (defect BP-6).

FR002: Charging Station Search

Priority	Jira reference	Story
High	SBP-2 / KP-15, 5 SP	The driver can search for charging stations and see which chargers are suitable and available.

Requirements

- **FR002.1** The system shall let the driver search by location (and optionally by time).
- **FR002.2** The system shall filter results by connector compatibility and charger availability.
- **FR002.3** The system shall show, for each result, the station, charger type, availability and current rate.

Acceptance criteria

- Search results contain only compatible chargers (defect BP-1).
- Chargers that are occupied or out of service are not shown as available.

FR003: Dynamic Power Rate Calculation

Priority	Jira reference	Story
Medium	SBP-3 / KP-16, 5 SP	The system calculates a charging rate that changes with defined conditions and shows it to the driver.

Requirements

- **FR003.1** The system shall calculate the charging rate using the pricing rules (for example demand, time of day and charger power level) (**assumed factors**).
- **FR003.2** The system shall display the applicable rate before the driver confirms a reservation.
- **FR003.3** The rate shown at confirmation shall be the rate used for billing the reservation.

Acceptance criteria

- For the same inputs the calculated rate is always the same and matches the pricing rules (defect BP-3).
- The displayed rate equals the stored rate on the reservation.

FR004: Reservation Cancellation

Priority	Jira reference	Story
Medium	SBP-4 / KP-17, 3 SP	When a user cancels a reservation, the system should release the slot and update its status to available so that other users can book it.

Requirements

- **FR004.1** The system shall let a driver cancel an active reservation.
- **FR004.2** The system shall set the reservation status to Cancelled.
- **FR004.3** The system shall release the reserved slot and update its status to Available.

Acceptance criteria

- After cancellation the reservation is no longer active (defect BP-4).
- The cancelled slot can be booked by another user immediately.

FR005: Charger Status Management

Priority	Jira reference	Story
High	SBP-5 / KP-18, 5 SP	The system keeps the status of every charger correct and synchronised across all functions.

Requirements

- **FR005.1** The system shall maintain the status of each charger: Available, Reserved, Occupied or Out of service.
- **FR005.2** The system shall update the status whenever a reservation is made, cancelled, released, started or completed.
- **FR005.3** The system shall synchronise availability so that search and reservation use the same current status.

Acceptance criteria

- A status change is visible in search results without manual refresh by the operator (defect BP-5).

3.3 Non-Functional Requirements

NFR001: No-Show Handling Performance

Priority	Jira reference	Category
High	SBP-6 / KP-19, 8 SP	Reliability and performance

- **NFR001.1** The system shall detect a no-show when the driver has not arrived or started charging within 10 minutes of the reservation start time.
- **NFR001.2** The system shall automatically release the socket after a no-show is detected and make the slot Available.
- **NFR001.3** The release shall happen within 1 minute of the 10-minute limit (**assumed tolerance**).
- **Acceptance:** a reservation with no arrival is released after 10 minutes (defect BP-7).

NFR002: System Performance and Scalability

Priority	Jira reference	Category
Medium	SBP-16 / KP-20, 12 SP	Performance and scalability

- **NFR002.1** Search and reservation requests shall respond within 3 seconds under normal load (**assumed target**).
- **NFR002.2** The system shall keep working correctly during peak load, with no double booking of a slot (**target number of concurrent users to be fixed by peak-load testing, assumed**).
- **NFR002.3** The system shall be able to handle more stations and users by adding resources, without redesign.
- **Acceptance:** peak-load test (SBP-29) and scalability validation (SBP-30) pass the agreed targets.

3.4 Other Quality Attributes (assumed)

Attribute	Requirement
Security	Only logged-in drivers can reserve or cancel; a driver can view and cancel only their own reservations; payment data is not stored by the system.
Usability	A new driver should be able to find a charger and reserve a slot without training.
Data integrity	A slot can have at most one active reservation at any time.

3.5 Reported Defects Linked to Requirements

These seven defects were logged in the Bug Tracking space. Each one is a way the above requirements can fail, so they are used as test targets.

Bug ID	Summary	Related requirement
BP-1	Charging station search returns incompatible chargers	FR001, FR002
BP-2	Reservation allows an unavailable charging slot	FR001
BP-3	Dynamic charging rate is calculated incorrectly	FR003
BP-4	Cancelled reservation remains active	FR004
BP-5	Charger availability status is not updated	FR005
BP-6	Payment succeeds but reservation is not confirmed	FR001
BP-7	No-show reservation does not release the socket after 10 minutes	NFR001

4. Requirements Traceability

Summary of how each requirement maps to Jira work. The full RTM with design and test case columns is kept in the Requirements Engineering folder.

Req ID	Requirement	Priority	Scrum / Kanban story	Tasks (Scrum / Kanban)	Defect
FR001	Compatible Charger Reservation	High	SBP-1 / KP-14	SBP-17, 18 / KP-21, 22	BP-1, 2, 6
FR002	Charging Station Search	High	SBP-2 / KP-15	SBP-19, 20 / KP-23, 24	BP-1
FR003	Dynamic Power Rate Calculation	Medium	SBP-3 / KP-16	SBP-21, 22 / KP-25, 26	BP-3
FR004	Reservation Cancellation	Medium	SBP-4 / KP-17	SBP-23, 24 / KP-27, 28	BP-4
FR005	Charger Status Management	High	SBP-5 / KP-18	SBP-25, 26 / KP-29, 30	BP-5

Req ID	Requirement	Priority	Scrum / Kanban story	Tasks (Scrum / Kanban)	Defect
NFR001	No-Show Handling Performance	High	SBP-6 / KP-19	SBP-27, 28 / KP-31, 32	BP-7
NFR002	System Performance and Scalability	Medium	SBP-16 / KP-20	SBP-29, 30 / KP-33, 34	-

5. Appendix A: Jira Work Item Mapping

Task	Scrum ID	Kanban ID	Parent requirement
Implement connector compatibility matching	SBP-17	KP-21	FR001
Implement reservation slot creation	SBP-18	KP-22	FR001
Implement charging station search	SBP-19	KP-23	FR002
Implement charger availability filtering	SBP-20	KP-24	FR002
Implement dynamic power rate calculation	SBP-21	KP-25	FR003
Display calculated charging rate	SBP-22	KP-26	FR003
Implement reservation cancellation	SBP-23	KP-27	FR004
Release cancelled reservation slot	SBP-24	KP-28	FR004
Implement charger status updates	SBP-25	KP-29	FR005
Synchronize charger availability	SBP-26	KP-30	FR005
Implement no-show detection	SBP-27	KP-31	NFR001
Implement automatic socket release	SBP-28	KP-32	NFR001
Perform peak-load performance testing	SBP-29	KP-33	NFR002
Validate system scalability	SBP-30	KP-34	NFR002

Example subtasks: SBP-31 / KP-35 Validate supported connector types; SBP-38 Mark cancelled slot as available; KP-40 Display applicable charging rate.

6. Appendix B: Revision History

Version	Date	Author	Change
1.0	02 Oct 2026	Shriya S R	First version